

Numerical Simulation of Blood Flow Through Microvascular Capillary Networks

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Abstract A numerical method is implemented for computing blood flow through a branching microvascular capillary network. The simulations follow the motion of individual red blood cells as they enter the network from an arterial entrance point with a specified tube hematocrit, while simultaneously updating the nodal capillary pressures. Poiseuille's law is used to describe flow in the capillary segments with an effective viscosity that depends on the number of cells residing inside each segment. The relative apparent viscosity is available from previous computational studies of individual red blood cell motion. Simulations are performed for a tree-like capillary network consisting of bifurcating segments. The results reveal that the probability of directional cell motion at a bifurcation (phase separation) may have an important effect on the statistical measures of the cell residence time and scattering of the tube hematocrit across the network. Blood cells act as regulators of the flow rate through the network branches by increasing the effective viscosity when the flow rate is high and decreasing the effective viscosity when the flow rate is low. Comparison with simulations based on conventional models of blood flow regarded as a continuum indicates that the latter underestimates the variance of the hematocrit across the vascular tree.

Keywords Blood flow · Capillary network · Microcirculation · Red blood cells

1. Introduction

The hydrodynamics and transport properties of blood flow in the microcirculation has been the subject of extensive laboratory and numerical investigations dating back to Poiseuille's seminar work on tube flow (Sutera and Skalak, 1993). An important goal has been the study of red blood cell motion and number density distribution expressed in terms of the discharge and tube hematocrit; the former is the ratio of the flow rate of the cells to the flow rate of whole blood, and the later is the volume fraction of the suspended cells. The ultimate objective is the prediction of metabolite and peripheral oxygen transport in health and disease (e.g., Popel, 1989; Pries et al., 1996). Krogh (1919)

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proposed an elementary microvascular model for skeletal muscle consisting of an array of evenly spaced capillaries. Oxygen is transported along the blood vessels and escapes into the tissue through the vascular walls where it diffuses and is consumed at a specified metabolic rate. Krogh's model was improved, enhanced, and elaborated by subsequent authors including Apelblat et al. (1974) and Blake and Gross (1982). Oxygen transport in different capillary arrangements pertinent to different types of tissue, and discrete models of oxygen delivery through the microvascular network of muscle, brain, and tumor tissue were developed by other authors (Beard and Bassingthwaite, 2001; Goldman and Popel, 2000; Secomb and Hsu, 1988, 1994; Hsu and Secomb, 1989; Secomb et al., 1993, 1995, 2000, 2004).

Studies of blood flow through microvascular networks have revealed significant variations in the red blood cell velocity and number density due to the diverse topology and geometry of the vascular bed, as discussed by Popel and Johnson (2005). In extreme cases, neoplastic vascular networks of solid tumors are devoid of red blood cells. The repartitioning of cells at a capillary bifurcation, described as phase separation, has been identified as an important parameter of the particulate hydrodynamics. In numerical simulations, the relationship between the discharge hematocrit of a daughter capillary vessel to that of a parent vessel is expressed by a bifurcation law. For example, Price and Skalak (1995) considered blood flow through the capillary network of a typical rat skeletal muscle in a transverse arteriole-collecting venule (TA-CV) module. The effective viscosity of blood in each capillary was determined by the capillary diameter and discharge hematocrit according to relations derived by Pries et al. (1990, 1992). The individual capillary discharge hematocrit was determined by a correlation (bifurcation law) at divergent bifurcations, and by requiring cell conservation at convergent bifurcations.

In the present work, a numerical method is implemented for describing blood flow in a vascular capillary network by tracking the motion of the individual red blood cells. The approach is novel in that the bifurcation law used by previous authors is replaced by a probability that a red blood cell follows one branching segment or another at a bifurcation. The probabilities can be deduced either from laboratory observations or from numerical simulations of the individual cell motion at a bifurcation (e.g., Popel, 1989). The flow in each capillary segment is governed by Poiseuille's law with an effective fluid viscosity that depends on the number of contained cells and capillary diameter using information extracted from cell-level hydrodynamics. Knowledge of the instantaneous effective viscosity circumvents the need for iterations, apart from those required for the solution of a linear system for the capillary nodal pressures. An attractive feature of the discrete approach is that the dynamics of the particulate flow can be described during a transient start-up period, and statistical features of the motion including scattering of the hematocrit can be studied at dynamic equilibrium.

The model of particulate capillary flow discussed in this paper builds on earlier models where the motion of individual cells or clumps of cells forming slugs was simulated by numerical methods. Schmid-Schönbein et al. (1980) were the first to consider the motion of discrete red and white blood cells in capillary networks. The hematocrit ratio, χ , and apparent intrinsic viscosity, K_T , were assumed to correlate with the ratio of an equivalent cell radius to the capillary radius according to relations derived for rigid spherical and biconcave particles. Computations were performed for a single capillary branch hosting up to sixteen cells, and for a modest size network replicating that observed in the rabbit momentum. It appears that the Heaviside function was used to determine the

cell motion at a bifurcation, resulting in some empty capillary segments. The simulations demonstrated that nonlinearities in the directional probability function is the source of significant nonuniformity in the tube hematocrit. Similar computations with red and white blood cells were performed by Fenton et al. (1985) and Furman and Olbricht (1985).

Slug models divide the fluid in each capillary into cylindrical elements that are large compared to the size of the red cells but small compared to the length of the capillaries (Kiani et al., 1994). A slug contains all cells entering an inflow vessel over a specified time increment. The length of each slug is equal to the average red blood cell velocity multiplied by the computational time interval. Slugs inside a capillary tube can have different tube hematocrit. The effective viscosity of blood inside each capillary is computed by averaging the slug contents. The stochastic nature of the red blood cell directional motion at a bifurcation distinguishes the present computations from the discrete cell and slug computations performed by previous authors.

The structure of capillary networks in the microcirculation strongly depends on the host organ and type of tissue. Because of its pronounced randomness, the vascular architecture of myocardial and tumor tissue has been described under the auspices of fractal geometry (e.g., Bassingthwaighe et al., 1994; Beard and Bassingthwaighe, 2000; Gazit et al., 1997). In numerical simulations, self-similar and completely randomly randomized structures with a specified fractal dimension or other statistical measures can be generated by invasion percolation branching and random growth techniques (e.g., Tsafnat et al., 2004). The capillary network model considered in our numerical studies consists of a cascading sequence of bifurcating capillary segments forming a randomized fractal-tree structure similar to that observed in the kidney vasculature (Karshafian et al., 2003). The construction involves several random parameters causing deviations from a perfectly ideal structure. Other capillary networks with arbitrary geometry and topology can be handled in a straightforward fashion.

2. Construction of the capillary network

Our model capillary network begins at an arterial entrance point defining the first end point of the first capillary segment in the xy plane with specified length, L_{c_1} , and radius, a_{c_1} . A sequence of bifurcations is then introduced to produce a branching network, as shown in Fig. 1(a). The length of the projection of each generated capillary segment in the xy plane is $L_c = \omega L'_c [1 + \epsilon_L (r - 0.5)]$, where L'_c is the length of the parental segment, ω is a specified contraction coefficient, ϵ_L is a specified dimensionless parameter, and r is a uniform deviate. The bifurcation angle in the xy plane is $\theta = \pm\theta_0 + \epsilon_\theta (r - 0.5)\pi$, where θ_0 is a specified mean semiangle and ϵ_θ is a specified dimensionless parameter. After a capillary bifurcation has been generated parallel to the xy plane, the second end point of each newly created capillary segment is displaced normal to the xy plane by the distance $z = L_{c_1} \epsilon_z (r - 0.5)$, where ϵ_z is a specified dimensionless parameter. The radius of each generated capillary segment is reduced geometrically with respect to that of the parent segment, so that the last bifurcation yields a specified minimum radius, $a_{c_{\min}}$.

In summary, the bifurcating capillary network is parametrized by the length and radii L_{c_1} , a_{c_1} , and $a_{c_{\min}}$, the dimensionless capillary length contract ratio ω , the dimensionless parameters ϵ_L , ϵ_θ , and ϵ_z determining the degree of randomness, and the number of bifurcations, m . When $\epsilon_L = 0$, $\epsilon_\theta = 0$, and $\epsilon_z = 0$, the construction is deterministic. The tree-like network is similar to that considered by Karshafian et al. (2003) in their study of

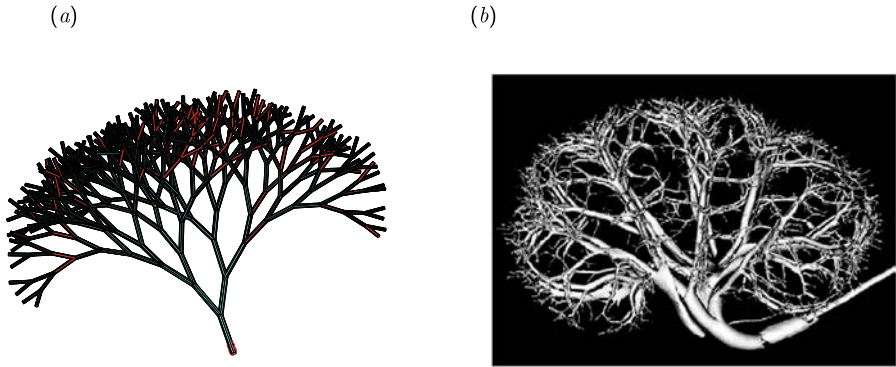


Fig. 1 (a) A tree-like capillary network model exhibiting random bifurcations and terminating after a specified number of bifurcations. The capillary radius decreases geometrically by a constant factor from the entrance point to the smallest capillaries around the periphery. (b) Surface rendering of rabbit kidney data acquired with high resolution computed tomography (micro-CT); the vasculature was perfused with an x-ray contrast agent through the renal artery; the horizontal field of view is 37 mm (Karshafian et al., 2003; image courtesy of M. Marxen, University of Toronto).

transit time kinetics in normal kidney tissue and in the abnormal neoplastic vasculature of solid tumors, as shown in Fig. 1(b). The authors demonstrated that simple manipulation of branching parameters yields vascular trees that mimic both normal kidney-like and abnormal tumor-like networks. In physiology, the smallest arterioles of the vascular network at the end of the capillary tree are connected to the venules of a conjugate venous network. Since the pressure drop is small across the venules, the venous pressure can be assumed constant at the terminal branches of the capillary tree, and the flow rate arises as part of the solution.

The numerical description of the capillary tree network borrows concepts from finite- and boundary-element implementations where the notion of element and node (branching) connectivity is of central importance. In the numerical construction, each straight capillary segment is subtended between two segment end nodes whose union defines a set of unique global nodes. Segment end nodes, global nodes, segment labels, and vice versa, are mapped through connectivity tables. Host capillary segments for each unique global node are identified, and neighbors are thereby deduced.

3. Capillary blood flow

Blood flow through a capillary segment of length L_c and radius a_c can be described by Poiseuille's law with an effective fluid viscosity, μ_{eff} , that depends on the pressure drop across the capillary, number of suspended cells, and capillary segment radius. The capillary flow rate is

$$Q = \frac{\Delta p_c \pi a_c^4}{8 \mu_{\text{eff}} L_c}, \quad (1)$$

where Δp_c is the pressure difference across the capillary end nodes. Due to entrance effects, imperfections, and deviations from the circular shape, Poiseuille's law should

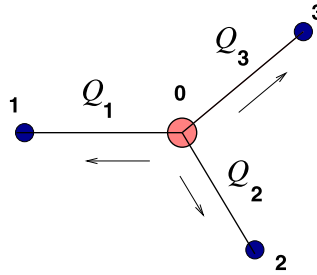


Fig. 2 Illustration of a capillary bifurcation. The flow rates are positive when fluid is driven away from the bifurcation. A mass balance yields a relation between the pressures at the four segment nodes.

be regarded as a scaling law involving effective geometrical and fluid parameters. For a particulate medium, the effective viscosity is interpreted as a proportionality coefficient relating flow rate to pressure drop.

Consider a capillary bifurcation at a node where one parental capillary segment bifurcates into two segments in three-dimensional space, as shown in Fig. 2. The local configuration involves four pressure nodes representing network junctions. To develop an iterative solution scheme for the junction pressures, we write a mass balance at the bifurcation node involving four pressures multiplied by coefficients playing the role of transfer functions, solve for the pressure at the central node, and iterate based on the derived expression. The result is the counterpart of the point Gauss–Seidel method (PGS) for solving sparse systems of linear equations originating from Laplace’s equation (e.g., Pozrikidis, 2008). With reference to the configuration depicted in Fig. 2, the mass balance requires $Q_1 + Q_2 + Q_3 = 0$, or

$$(P_0 - P_1) \frac{a_{c_1}^4}{\mu_{\text{eff}_1} L_{c_1}} + (P_0 - P_2) \frac{a_{c_2}^4}{\mu_{\text{eff}_2} L_{c_2}} + (P_0 - P_3) \frac{a_{c_3}^4}{\mu_{\text{eff}_3} L_{c_3}} = 0 \quad (2)$$

(Lipowsky and Zweifach, 1974). The numerical procedure involves guessing P_1 , P_2 , and P_3 , and solving (2) for P_0 to obtain an update. The pressure is sequentially updated at all unique global nodes, and the iterations continue until the maximum correction falls below a specified threshold set to 10^{-6} times the inlet pressure. The computations converge after a moderate number of iterations for all configurations considered.

4. Red blood cell motion and effective apparent viscosity

The effective blood viscosity in capillary flow depends on the capillary radius, cell number density, and flow rate according to the Fåhræus–Lindqvist effect. Consider the axisymmetric motion of a file of tightly-fitting red blood cells through a narrow capillary. The apparent intrinsic viscosity of the particulate fluid, K_T , is defined by the equation

$$\mu_{\text{eff}} \equiv \mu(1 + H_T K_T), \quad (3)$$

where μ is the plasma viscosity and H_T is the tube hematocrit defined as the volume fraction of the suspended phase (e.g., Secomb et al., 1986). For a periodic file of evenly spaced

cells separated by the distance L_{rbc} inside a capillary tube of radius a_c and length L_c ,

$$H_T = \frac{V_{rbc}}{\pi a_c^2 L_{rbc}} = n \frac{V_{rbc}}{\pi a_c^2 L_c}, \quad (4)$$

where V_{rbc} is the individual cell volume, and $n = L_c/L_{rbc}$ is the number of cells in the tube. Thus,

$$\mu_{\text{eff}} \equiv \mu \left(1 + n \frac{V_{rbc}}{\pi a_c^2 L_c} K_T \right). \quad (5)$$

This expression can be used to compute the effective viscosity in a capillary tube in terms of the number of cell residing inside the tube, n , the length and radius of the tube L_c and a_c , and the apparent intrinsic viscosity, K_T .

The discharge hematocrit, H_D , is defined as the ratio of the volumetric flow rate of the cells averaged over a long period of time, Q_{rbc} , to the flow rate of the whole suspension, Q . The mean velocity of the suspension is $U_m \equiv Q/\pi a_c^2$. If the cells travel along the tube as rigid bodies with velocity U_{rbc} , then, by definition, $Q = U_{rbc} V_{rbc}/L_{rbc}$, and

$$H_D \equiv \frac{Q_{rbc}}{Q} = \frac{U_{rbc} V_{rbc}}{U_m \pi a_c^2 L_{rbc}}. \quad (6)$$

Thus, the ratio of the tube hematocrit to the discharge hematocrit is (Sutera et al., 1970)

$$\chi \equiv \frac{H_T}{H_D} = \frac{U_m}{U_{rbc}}. \quad (7)$$

A lower bound on χ can be derived by setting the cell velocity equal to the Poiseuille-flow centerline velocity, $U_{rbc} = 2U_m$, finding $\chi = 0.5$. Laboratory observation and theoretical models have confirmed that χ is less than unity for a broad range of conditions (Pries et al., 1992). The reduction of χ from the value of unity is known as the Fåhræus effect.

Secomb et al. (1986) presented computational results illustrating the dependence of the hematocrit ratio, χ , and apparent intrinsic viscosity, K_T , on the cell velocity for widely separated cells. Results presented in their Fig. 7 show that χ varies between 0.7 and 1.0 over a broad range of cell velocities, from 0.001 to 1 cm/s, and over a broad range of capillary radii, from 3.4 to 8 μm . For well-separated cells, K_T depends on a_c and U_m . Results presented in Fig. 6 of Secomb et al. (1986) show that the apparent intrinsic viscosity, K_T , is nearly independent of the cell velocity in the range 0.1 to 1 cm/s, and sharply increases for smaller cell velocities. For $U_{rbc} \simeq 1$ cm/s, K_T increases from nearly 1.2 to nearly 2.4 as the tube radius decreases from 6.0 to 3.4 μm . Pozrikidis (2005) performed boundary-integral simulations of the axisymmetric motion of a periodic file of red blood cells and confirmed these predictions.

Using the definition of χ , we may express the effective viscosity of blood in capillary flow as

$$\mu_{\text{eff}} \equiv \mu(1 + \chi H_D K_T), \quad (8)$$

with the understanding that χ and K_T are implicit functions of H_D and capillary tube radius, a_c , and vice versa. Pries et al. (1990) proposed the correlation

$$\chi = H_D + (1 - H_D)(1 + 1.7e^{-0.7a_c} - 0.6e^{-0.02a_c}). \quad (9)$$

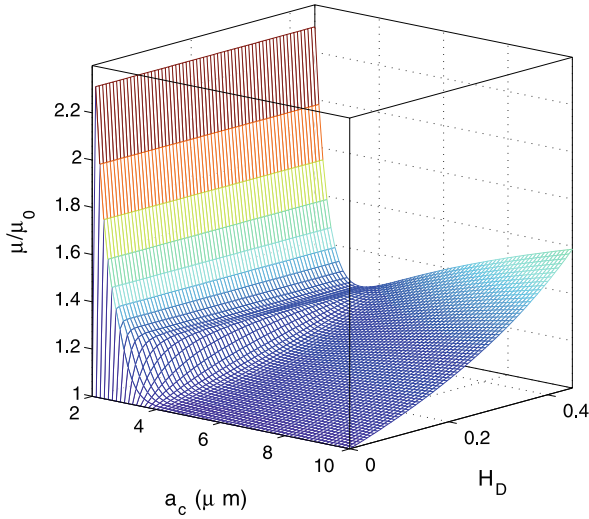


Fig. 3 Dependence of the effective blood viscosity in tube flow on the capillary radius, a_c , and discharge hematocrit, H_D .

Pries et al. (1992) proposed a correlation for the blood viscosity in a capillary of radius a_c measured in μm ,

$$\mu_{\text{eff}} = \mu \left[1 + (\mu_{45} - 1) \frac{(1 - H_D)^C - 1}{(1 - 0.45)^C - 1} \right], \quad (10)$$

where

$$C = (0.8 + e^{-0.15a_c}) \left(-1 + \frac{1}{f} \right) + \frac{1}{f}, \quad (11)$$

$f = 1 + 0.1(a_c/5)^{12}$, and μ_{45} is the relative blood viscosity at $H_D = 0.45$ given by

$$\mu_{45} = 220e^{-2.6a_c} + 3.2 - 2.44e^{-0.06(2a_c)^{0.645}}. \quad (12)$$

A graph of the relative blood viscosity according to this correlation is shown in Fig. 3. When $H_D = 0$, the viscosity is constant and equal to the plasma viscosity, independent of the tube radius. At nonzero hematocrits, a sharp increase is observed when the capillary radius becomes approximately $2 \mu\text{m}$ due to the inability of the red blood cells to squeeze through the capillary vessels.

In network models of biphasic blood flow, a discharge hematocrit is assigned to the each capillary segment at steady state. Conservation of red blood cell count at the bifurcation illustrated in Fig. 2 requires

$$Q_1 H_{D_1} + Q_2 H_{D_2} + Q_3 H_{D_3} = 0. \quad (13)$$

Substituting (3) in the whole blood and cell balance Eqs. (2) and (13), we find

$$\frac{a_{c_1}^4 (P_0 - P_1)}{L_{c_1} (1 + \chi_1 H_{D_1} K_{T_1})} + \frac{a_{c_2}^4 (P_0 - P_2)}{L_{c_2} (1 + \chi_2 H_{D_2} K_{T_2})} + \frac{a_{c_3}^4 (P_0 - P_3)}{L_{c_3} (1 + \chi_3 H_{D_3} K_{T_3})} = 0 \quad (14)$$

and

$$\frac{a_{c_1}^4 (P_0 - P_1) H_{D_1}}{L_{c_1} (1 + \chi_1 H_{D_1} K_{T_1})} + \frac{a_{c_2}^4 (P_0 - P_2) H_{D_2}}{L_{c_2} (1 + \chi_2 H_{D_2} K_{T_2})} + \frac{a_{c_3}^4 (P_0 - P_3) H_{D_3}}{L_{c_3} (1 + \chi_3 H_{D_3} K_{T_3})} = 0. \quad (15)$$

Consider a model configuration where the peripheral pressures, P_1 , P_2 , and P_3 , as well as the inlet hematocrit, H_{D_3} , are specified. Equations (14) and (15) provide us with a system of two equations for three unknowns, P_0 , H_{D_1} , and H_{D_2} . To obtain a unique solution, an additional relation describing the physics of the particulate flow at a bifurcation is required. In the discrete model pursued in this work, this information is mediated by a probability determining the direction of the individual cell motion at a bifurcation.

5. Cell injection, removal, and motion at a bifurcation

In the numerical simulation, cells are injected into the first arterial segment at a sequence of time instants. The injection time interval, Δt_{inj} , is determined by the specified inlet tube hematocrit, $H_{T,\text{in}}$, and the computed current first arterial segment cell velocity, U_{rbc} , according to the relation $\Delta t_{\text{inj}} = d/U_{rbc}$, where $d = V_{rbc}/(\pi a_{c,1}^2 H_{T,\text{in}})$ is the cell separation. The position of the cells is advanced over a time interval $\Delta t = \frac{1}{r} \Delta t_{\text{inj}}$, where r is a specified integer; thus, injection occurs approximately every r time steps. The value of r must be sufficiently large so that the cells travel by a distance that is a small fraction of the capillary length during each time step. However, a high value of r considerably increases the cost of the simulation by requiring a large number of time steps to reach dynamic equilibrium. Numerical experimentation has shown that the value $r = 8$ used in most simulations strikes a good balance between consistency and efficiency. The number of cells in each capillary segment is calculated after each time step by updating a segment bin count. When a cell exits or enters a capillary segment, the bin count is reduced or increased by one unit. The effective viscosity of the fluid inside each capillary segment is computed at each time step based on the discharge hematocrit, and the nodal pressures of the capillary network are calculated by point Gauss–Seidel iterations. When a cell exits an outermost capillary segment, it is tagged and removed from the simulation.

An important feature of the numerical model is the stochastic nature of the directional cell motion at a bifurcation. When a cell exits a nonterminal capillary segment, it enters one of two descendant bifurcating segments labeled 1 and 2, with respective probabilities ϕ_1 and ϕ_2 , where $\phi_1 + \phi_2 = 1$. In the numerical implementation, the direction of the cell motion is determined by a random number generator producing uniform deviates in the range $[0, 1]$. If a random number is less than ϕ_1 , a cell enters the first capillary; otherwise, the cell enters the second capillary. The probabilities ϕ_1 and ϕ_2 depend on the capillary radii, branching angles, and flow rates. Because cells in large blood vessels crudely follow the streamlines of the bulk flow, the probabilities are approximately equal to the fractional flow rates, $\phi_i \simeq \psi_i \equiv Q_i/(Q_1 + Q_2)$. Cells in small vessels tend to follow the direction of the highest flow rate, and $\psi_i \simeq H(2Q_i - Q_1 - Q_2)$, where H is the Heaviside function.

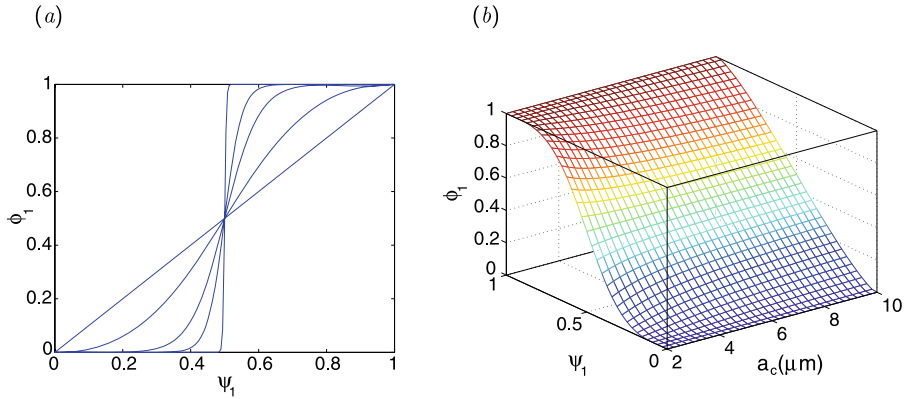


Fig. 4 Dependence of the probability of directional cell motion at a bifurcation, ϕ_1 , on the fractional flow rate according to (a) Eq. (16) for $q = 1$ (straight line), 2, 5, 10, and 100, and (b) Eq. (23) with $a_1 = a_2$ and inlet tube hematocrit $H_{D3} = 0.45$.

Fung (1973) and Schmid-Schönbein et al. (1980) discuss the dependence of the bifurcation probability function on the fractional flow rate and eccentric cell position over the capillary cross section. If the directional probabilities depend only on the fractional flow rates, $\phi_1 = f(\psi_1)$ and $\phi_2 = f(1 - \psi_1)$, where the function f satisfies $f(\psi_1) + f(1 - \psi_1) = 1$. Klitzman and Johnson (1982) proposed the more general relation

$$\phi_i = \frac{\psi_i^q}{\psi_i^q + (1 - \psi_i)^q}, \quad (16)$$

where q is an adjustable parameter. The linear dependence on the fractional flow rate corresponds to $q = 1$, and the Heaviside function arises in the limit as q tends to infinity. Figure 4(a) shows a graph of ϕ_1 plotted against ψ_1 for several values of q . As q increases, the probability takes a sigmoidal shape and eventually develops a discontinuity at the midpoint where “winner takes all.”

Now, we consider flow through a network at steady state. When a blood stream separates into two streams at a bifurcation, the tube and discharge hematocrits are redefined in a process that is sometimes called phase separation. A mass balance at a bifurcation requires $Q_1 + Q_2 + Q_3 = 0$, where Q_i is the whole blood flow rate, and the subscripts label the three capillaries involved, as illustrated in Fig. 2. Conservation of red blood cells at a bifurcation requires $Q_1^{rbc} + Q_2^{rbc} + Q_3^{rbc} = 0$, where Q^{rbc} is the erythrocyte flow rate. We assume that Q_1 , Q_2 , Q_1^{rbc} , and Q_2^{rbc} are positive corresponding to motion away from the bifurcation, whereas Q_3 and Q_3^{rbc} are negative corresponding to motion toward the bifurcation, and define the positive ratios

$$\psi_1 \equiv -\frac{Q_1}{Q_3}, \quad \phi_1 \equiv -\frac{Q_1^{rbc}}{Q_3^{rbc}}, \quad \psi_2 \equiv -\frac{Q_2}{Q_3}, \quad \phi_2 \equiv -\frac{Q_2^{rbc}}{Q_3^{rbc}}. \quad (17)$$

The ratios ϕ_1 and ϕ_2 are the probability of directional cell motion discussed earlier in this section. Conservation of whole blood requires $\psi_1 + \psi_2 = 1$ and conservation of red blood

cells requires $\phi_1 + \phi_2 = 1$. In terms of the discharge hematocrits, $H_{D_i} \equiv Q_i^{rbc}/Q_i$, we obtain

$$H_{D_1}\psi_1 + H_{D_2}\psi_2 = H_{D_3}. \quad (18)$$

Hydrodynamics provides us with a relation for the probability ratio

$$\phi_{12} \equiv \frac{\phi_1}{\phi_2} = \frac{H_{D_1}}{H_{D_2}} \frac{\psi_1}{\psi_2} = \mathcal{F}_1\left(a_3, H_{D_3}; \psi_2, \frac{a_1}{a_2}\right), \quad (19)$$

where a_i are the vessel radii, and the fractional flow rates of the particulate phase are given by

$$\phi_1 = \frac{\phi_{12}}{1 + \phi_{12}}, \quad \phi_2 = \frac{1}{1 + \phi_{12}}. \quad (20)$$

Symmetry requires

$$\mathcal{F}_1\left(a_3, H_{D_3}; \psi_1, \frac{a_1}{a_2}\right) \cdot \mathcal{F}_1\left(a_3, H_{D_3}; 1 - \psi_1, \frac{a_2}{a_1}\right) = 1. \quad (21)$$

Substituting (19) in (18) and rearranging, we find

$$H_{D_1} = \frac{\mathcal{F}_1}{\mathcal{F}_1 + 1} \frac{H_{D_3}}{\psi_1}, \quad H_{D_2} = \frac{1}{\mathcal{F}_1 + 1} \frac{H_{D_3}}{\psi_2}. \quad (22)$$

Given the tube radii, the inlet hematocrit, H_{D_3} , and the fractional whole blood flow rates, ψ_1 and ψ_2 , these equations can be used to compute the discharge hematocrits of daughter tubes. Pries et al. (1990) proposed the empirical relation

$$\mathcal{F}_1 = \left(\frac{a_1}{a_2}\right)^{-3.48/a_3} \cdot \left(\frac{\psi_1 - X_0}{1 - \psi_1 - X_0}\right)^q, \quad (23)$$

where $q = 1 + 3.49(1 - H_{D_3})/a_3$, $X_0 = 0.2/a_3$, and the radii, a_i , are measured in μm . For $X_0 = 0$ and $a_1 = a_2$, relation (23) reduces to (16). Figure 4(b) shows a graph of ϕ_1 plotted against ψ_1 for $a_1 = a_2$ and $H_{D_3} = 0.45$. Comparison with the curves shown in Fig. 4(a) shows that the exponent q increases from 1 at large capillaries to a value roughly equal to 5 at small capillaries.

6. Results and discussion

In the numerical simulations, the red blood cell velocity is set to 1.25 times the mean fluid velocity in the host capillary segment, corresponding to a constant hematocrit ratio, $\chi = 0.8$. The apparent intrinsic viscosity, K_T , is assumed to increase linearly from 1.2 to 2.4 as the tube radius decreases from 6.0 to 3.4 μm . Adoption of these simple forms is reasonable in light of important approximations made by neglecting the undulating shape and noncircular cross-section of capillaries, hydrodynamic cell interactions, and the presence of the endothelial surface layer (e.g., Secomb et al., 2001, 2002), as well

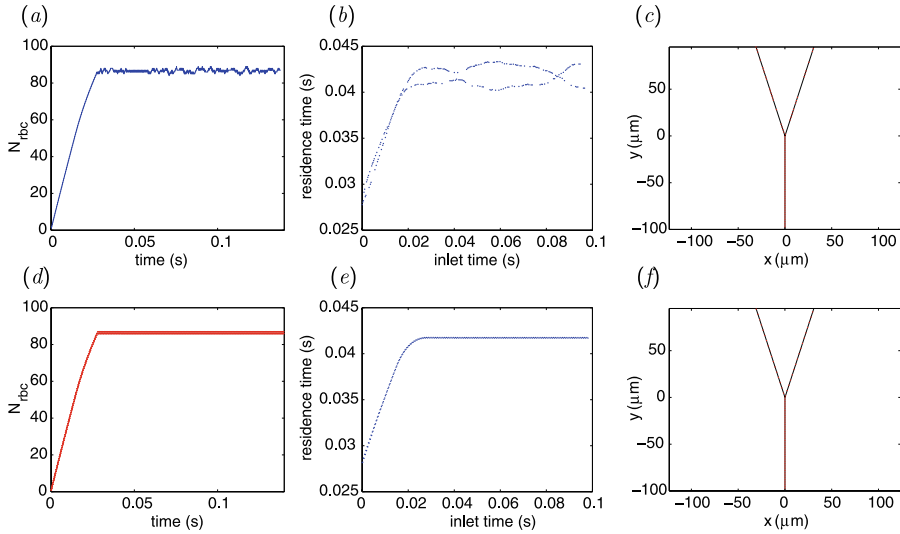


Fig. 5 Transient population, cell residence time, and instantaneous cell distribution in a symmetric bifurcation at the end of a simulation for probability exponent (a)–(c) $q = 1$ and (d)–(f) $q = \infty$. In frames (c), (f), the red blood cells are shown as red dots. (Color figure online.)

as by simplifying the physics of the cell motion at a bifurcation. The plasma viscosity is taken to be $\mu = 1.85 \text{ cp} = 1.85 \times 10^{-2} \text{ g/(cm s)}$, the radius and length of the inlet capillary segment are set to $6 \text{ }\mu\text{m}$ and $L_{c,1} = 100 \text{ }\mu\text{m}$, and the radius of the outermost capillary segments at the periphery of the network is set to $3.4 \text{ }\mu\text{m}$. The aspect ratio of the capillaries is typical of that encountered in the neoplastic vasculature of solid tumors (Less et al., 1991). The arterial pressure is set to $p_a = 10 \text{ mm Hg}$, and the venous pressure is set to zero.

A series of simulations was conducted to investigate the effect of the exponent q defined in the bifurcation law (16) on the statistics of the microvascular flow in a large network. To prepare the ground for the interpretation of simulations for large networks, in Fig. 5, we present results for a single symmetric bifurcation in the xy plane with branching semiangle $\theta_0 = \pi/10$, capillary segment contraction ratio $\omega = 0.9$, and vanishing random amplitudes, $\epsilon_L = 0$, $\epsilon_\theta = 0$, $\epsilon_z = 0$. The inlet tube hematocrit is $H_{T,in} = 0.35$; since $\chi = 1.25$, the inlet discharge hematocrit is $H_{D,in} = 0.4375$. Figures 5(a)–(c) correspond to probability exponent $q = 1$ where the direction of a cell at the bifurcation is determined by the fractional flow rate, and Figs. 5(d)–(f) correspond to probability exponent $q = \infty$ where the cells favor the segment with the largest flow rate. Although the physiological relevance of the Heaviside function associated with the choice $q = \infty$ could be questioned, consideration of this limit allows us to clearly illustrate the significance of phase separation. The results show that the number of cells inside the network increases and reaches a plateau as the cells start exiting the network. The cell residence time exhibits pronounced fluctuations when $q = 1$, and is a smooth function of time when $q = \infty$. These results indicate that the probability of directional motion of the cells at a bifurcation can appreciably affect the microphysics of the particulate flow.

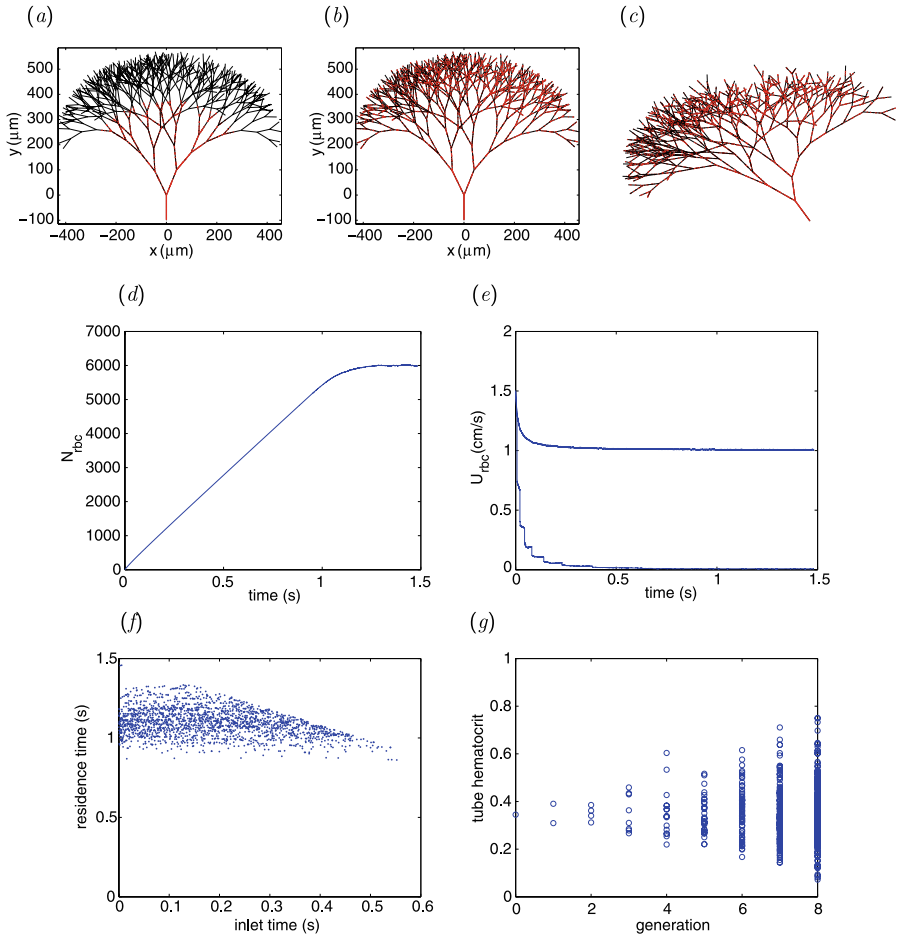


Fig. 6 (a), (b) Transient population of red blood cells in a capillary network with $m = 8$ generations of branching bifurcations and bifurcation exponent $q = 1$, at two different times; the red blood cells are shown as red dots. (c) Three-dimensional perspective of the network at the end of the simulation. (d) Number of cells inside the network, (e) evolution of the maximum and minimum cell velocity, (f) residence time of red blood cells, and (g) tube hematocrit of the capillary branches at the end of the simulation. (Color figure online.)

Figures 6(a)–(c) illustrates the transient population of a network with $m = 8$ generations of branching bifurcations, mean branching semiangle $\theta_0 = \pi/10$, capillary segment contraction ratio $\omega = 0.9$, and random amplitudes $\epsilon_L = 0.2$, $\epsilon_\theta = 0.1$, and $\epsilon_z = 0.5$. The inlet tube hematocrit is $H_{T,in} = 0.35$, corresponding to inlet discharge hematocrit $H_{D,in} = 0.4375$, and the bifurcation exponent is $q = 1$. The capillary network is drawn with solid lines and the red blood cells are shown as red dots. As red blood cells start entering the network, they are channeled through the capillary tree forming a radial front. Figures 6(a), (b) shows the projection of the network in the xy plane at two different times, and Fig. 6(c) shows a three-dimensional perspective of the populated network at

the end of the simulation. Variations in the number of cells in each capillary segment of each generation are discernible. This simulation involves 64,000 time steps and consumes several hours of CPU time on a high-end workstation. The computations are terminated when dynamic equilibrium is established, indicated either by the number of red blood cells residing inside the network, or by the number of cells that have exited the network.

Figure 6(d) shows a graph of the number of cells residing inside the network from the beginning to the end of the simulation, up to time approximately 1.5 s. The graph rises from zero at the origin of computational time to a well-defined mean value when dynamic equilibrium is established after an approximate transient period of 1.0 s. The number of cells inside the network exhibits small fluctuations thereafter. Nearly 6,000 red blood cells have entered the network during the transient startup period. In a perfectly deterministic network with zero randomness in the length and radius of the capillary segments, $\epsilon_L = 0$ and $\epsilon_\theta = 0$, the number of cells inside the network would increase linearly from zero at the origin of computational time to a constant value at the cell residence time, and will remain constant thereafter.

Figure 6(e) shows a graph of the maximum and minimum red blood cell velocity. The former occurs at the entrance point of the network, and the latter occurs at the front of the invading red blood cells. As the cells penetrate successive branches of the vasculature, the minimum velocity exhibits a stepwise decrease. The maximum velocity monotonically decreases from an initial high value corresponding to plasma flow to a lower value due to the increased network viscosity associated with the red blood cells. At dynamic equilibrium, the cell velocity varies in the approximate range 0.01 to 1 cm/s. Due to the assumption of constant hematocrit ratio, χ , the capillary mean velocity varies in a comparable range.

Figure 6(f) shows a graph of the cell residence time plotted against the inlet time for cells that have exited the network by the end of the simulation. All cells that have entered the network approximately up to time 0.2 s have exited the network, while only a fraction of cells that have entered the network at later times have exited the network, yielding a sloped right edge. The results reveal a significant variation in residence times, from 0.9 to 1.4 s. The mean residence time is approximately equal to the time required for the system to reach dynamic equilibrium, in agreement with intuition. Estimates for the mean residence time are useful for computing the amount of oxygen unloaded from the cells through the walls of the vasculature by diffusion.

Figure 6(g) illustrates the distribution of the tube hematocrit in the capillary branches labelled by their generation index, 0–8, at the end of the simulation. The tube hematocrit shown was computed by dividing the total cell volume inside each capillary by the capillary volume. The inlet tube hematocrit is equal to that specified by the inlet condition, $H_{T,in} = 0.35$. Significant scattering is observed for high-generation capillaries around the inlet value. Some capillary branches are nearly devoid of cells, whereas other branches are heavily populated. However, there is no bias as to what branches are loaded with cells, and a time average will reveal that each capillary in a generation hosts approximately the same number of cells over a long period of time. The pronounced variation in the cell residence time and tube hematocrit is due partly to the branching structure of the capillary network combined with the randomness of the directional cell motion at a bifurcation, and partly to the effect of the cells on the effective viscosity of the suspension in each capillary segment. The important effect of the viscosity contributed by the cells will be demonstrated later in our discussion.

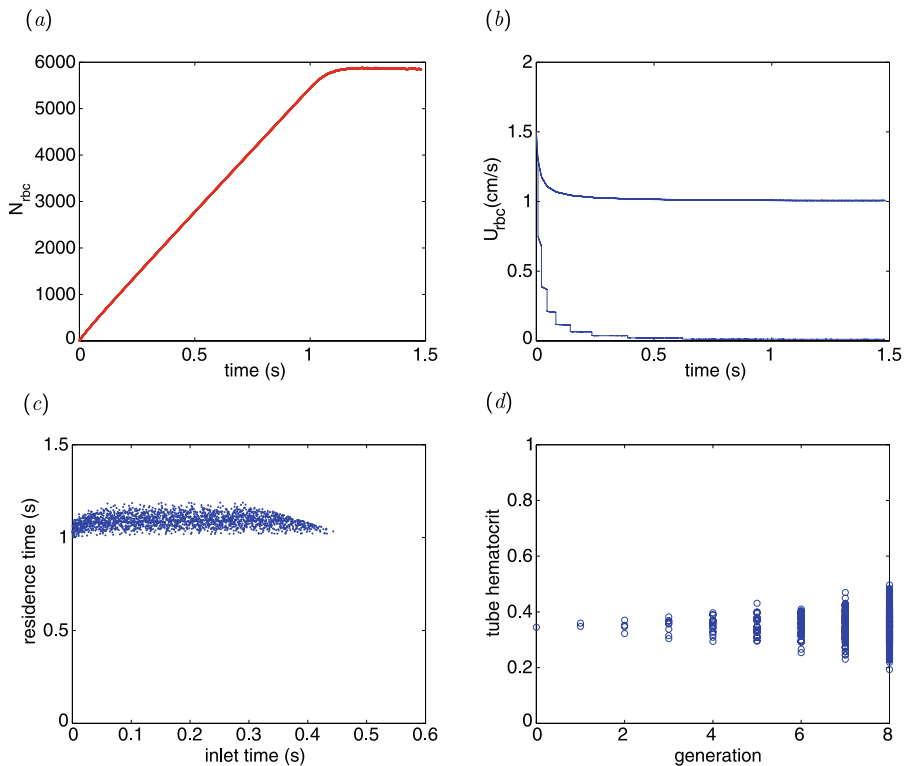


Fig. 7 Same as Figs. 6(d)–(g), but for bifurcation probability exponent $q = \infty$. In the simulation, the cells follow the direction of maximum flow rate at a bifurcation.

Figure 7 duplicates the results shown in Fig. 6 for $q = \infty$, where the cells follow the direction of maximum flow rate at a bifurcation. This high value of q was selected to magnify the effect of the cell motion when a blood stream divides. We see that the probability of directional motion has a minor effect on the number of cells residing inside the network and on the maximum and minimum cell velocities. In contrast, the bifurcation law has a strong effect on the scattering of the cell residence times and tube hematocrit. Both are reduced as q increases and the cells tend to follow the path of highest flow rate. These results are consistent with those shown in Fig. 5 for a single bifurcation.

Figure 8 shows results corresponding to those shown in Fig. 6, but for half the inlet tube hematocrit, $H_{T,in} = 0.175$. In both simulations, the cell motion at a bifurcation is determined by the fractional flow rate, $q = 1$. When $H_{T,in} = 0.175$, dynamical equilibrium is established after approximately 0.7 s, and the number of cells residing in the network is about half that for $H_{T,in} = 0.35$. Thus, the number of cells residing inside the network approximately scales with the inlet tube or discharge hematocrit. The cell residence time and distribution of tube hematocrit at the end of the simulation exhibit significant scattering. The mean residence time is smaller than that seen in the case of the high hematocrit due to the lower population of red blood cells and concomitant increased mobility of the network.

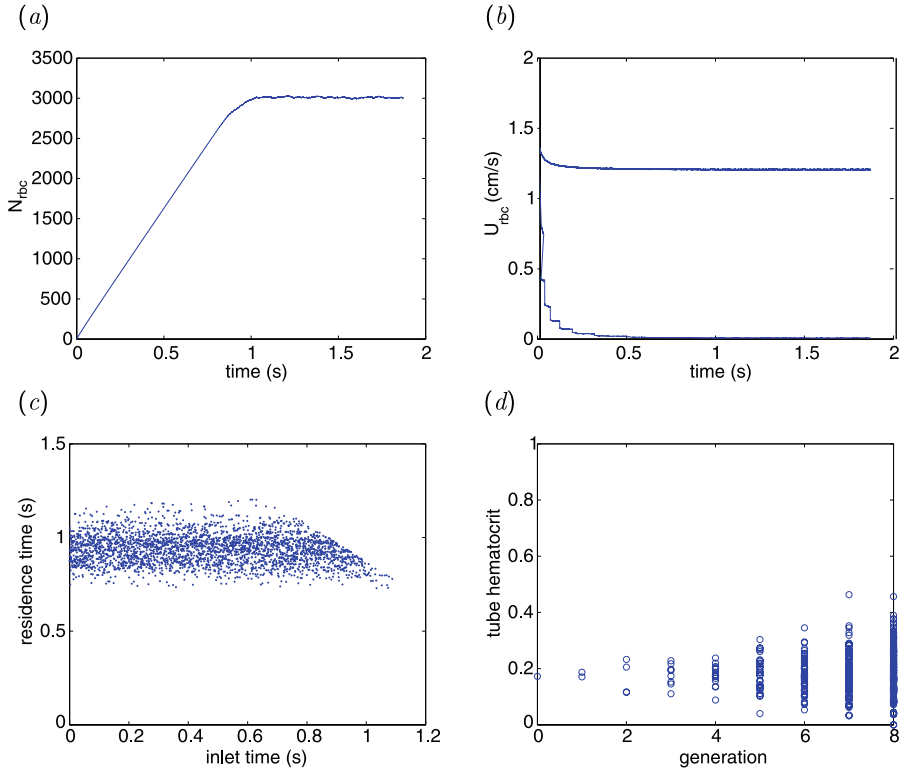


Fig. 8 Same as Figs. 6(d)–(g), for half the inlet tube hematocrit, $H_{T,in} = 0.175$.

Figure 9 shows results corresponding to those shown in Fig. 8, but for $q = \infty$. As in the case of the higher hematocrit, the number of cells residing inside the network at steady state is insensitive to the probability of cell motion at the bifurcations. Increasing q reduces the scattering of the cell residence time and tube hematocrit distribution across the network. These trends appear to be independent of the inlet hematocrit.

The numerical results suggest that cells act as internal equalizers of the flow rate across the network branches, by increasing the effective viscosity when the flow rate of a capillary is high and decreasing the effective viscosity when the flow rate is low. The action can be described as an adaptive internal control parametrized by the value of the exponent q , which is determined by the hydrodynamics of the cell motion at a bifurcation. A direct visual impression of the effect of q on the cell distribution across the network is presented in Fig. 10, showing snapshots of the cell distribution for a low inlet tube hematocrit $H_{T,in} = 0.0437$ and $q = 1$ or $q = \infty$. The ability of the cells to regulate the flow diminishes as the hematocrit, and thus the number of resident cells is reduced. Since the range of the exponent q in physiological networks with large and moderate capillary diameter ranges roughly between 1 and 5, the occurrence of flow regulation and self-sustained dynamics must be accepted with some reservation (Carr and Lacoïn, 2000; Carr et al., 2005).

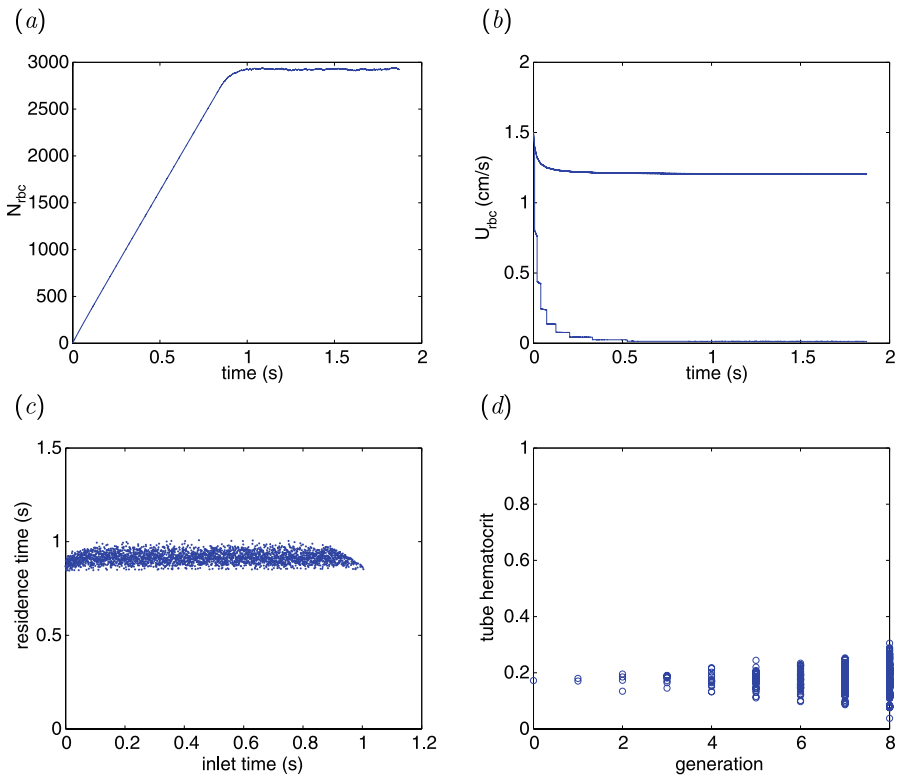


Fig. 9 Same as Fig. 8, but for probability exponent $q = \infty$.

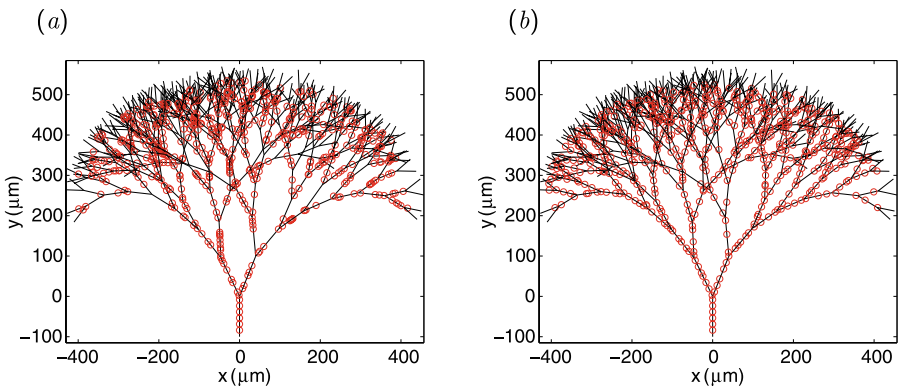


Fig. 10 Snapshots of cell distribution for inlet tube hematocrit $H_{T,in} = 0.0437$ and (a) $q = 1$ or (b) $q = \infty$. The cell distribution is more uniform on the right.

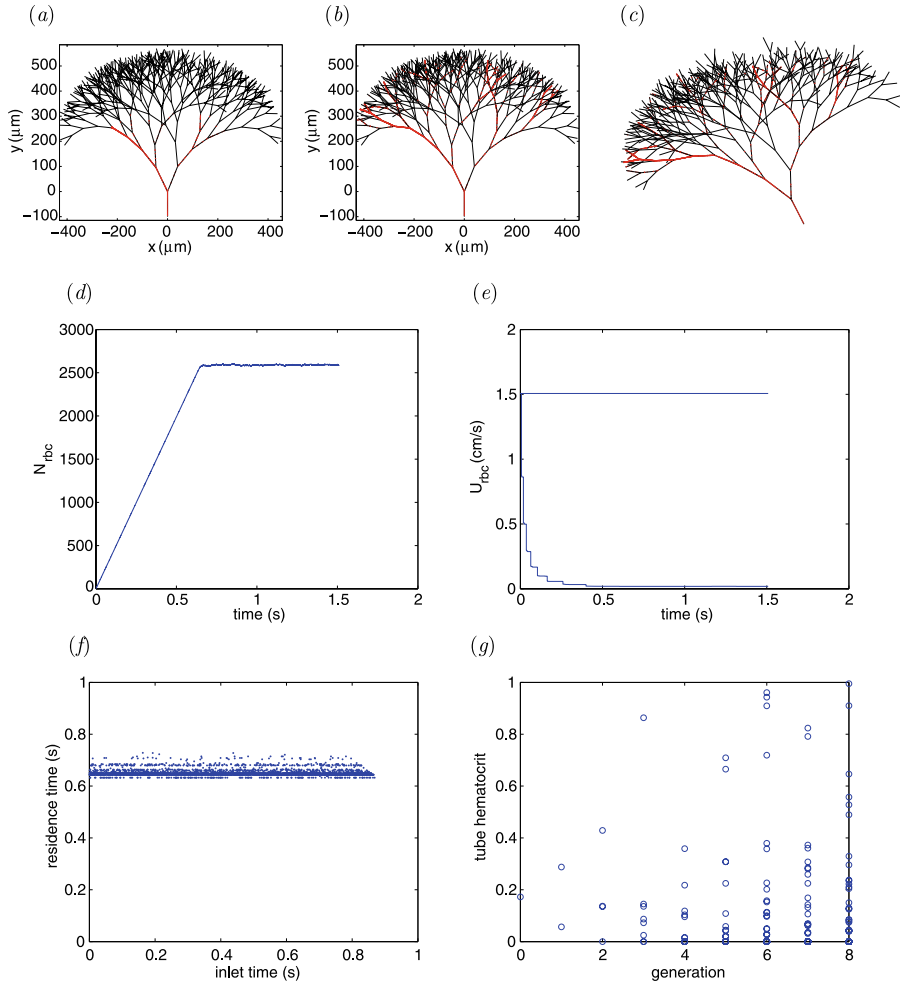


Fig. 11 Same as Fig. 8 but for constant viscosity and a higher probability exponent, $q = 100$.

To further demonstrate the importance of the fluid viscosity due to the cells on the cell motion through the network mediated by the value of the intrinsic apparent viscosity K_T , we discuss results of simulations for $H_{T,\text{inp}} = 0.175$ where the fluid viscosity is constant and equal to the plasma viscosity, corresponding to $K_T = 0$. For $q = 1$, we obtain smaller residence times than those shown in Fig. 8, a comparable number of resident cells, and a similar degree of scattering in the tube hematocrit at dynamic equilibrium. In the simulation shown in Fig. 11, the bifurcation exponent is set to $q = 100$. Thus, the overwhelming majority of cells follow the path of highest flow rate at each bifurcation, as shown in Fig. 6. The results demonstrate a possible pathological response, where the cells follow only a few capillary paths from the arterial entrance point to the venous exit points. Physically, because the cells have no effect on the effective viscosity of the suspension across a few initial bifurcations, cell-free paths develop. In fact, the standard implementation of

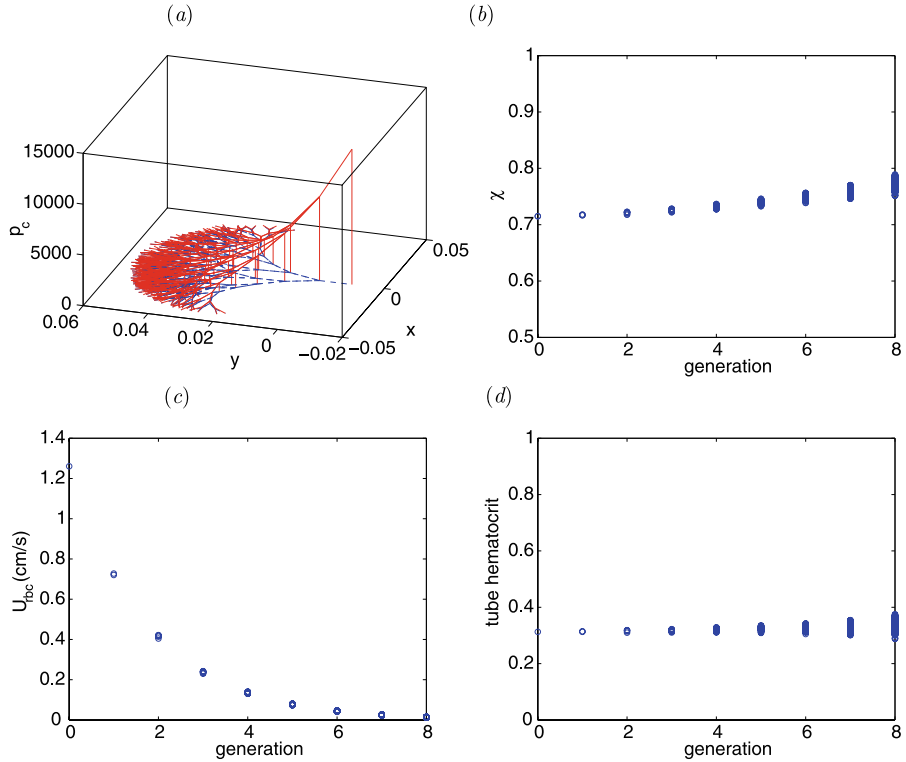


Fig. 12 Predictions of the continuum model corresponding to the simulations shown in Fig. 6. (a) Distribution of the capillary pressure measured in g/cm s^2 on the projection of the capillary network in the xy plane; Distribution of the (b) hematocrit ratio, (c) mean cell velocity, and (d) tube hematocrit across the network.

the numerical method fails in the absence of a restriction on the maximum number of cells residing in a capillary, as unphysical capillary tube hematocrits that are higher than unity arise. Setting $q = \infty$ yields a solitary cell path with a constant residence time reminiscent of invasion percolation. We conclude that low values of K_T coupled with high values of q lead to penetration pathways and necrotic regimes.

It is of interest to compare the predictions of the discrete blood flow model implemented in this work with the predictions of conventional network models where blood is regarded as a continuum. An iterative method was implemented to solve the continuum problem according to the following steps: the inlet discharge hematocrit is specified; guesses are made for the network capillary discharge hematocrits; the effective blood viscosity in each capillary tube is computed using the empirical relation (10); the capillary pressures and flow rates are found by Gauss–Seidel iterations as in the discrete formulation; the network capillary discharge hematocrits are updated using the empirical relation (22); the process is then repeated until convergence. The tube hematocrit is computed a posteriori from the discharge hematocrit using (9).

Figure 12 shows the predictions of the continuum model for vascular geometry and pressure conditions identical to those corresponding to Fig. 6, and inlet discharge hematocrit

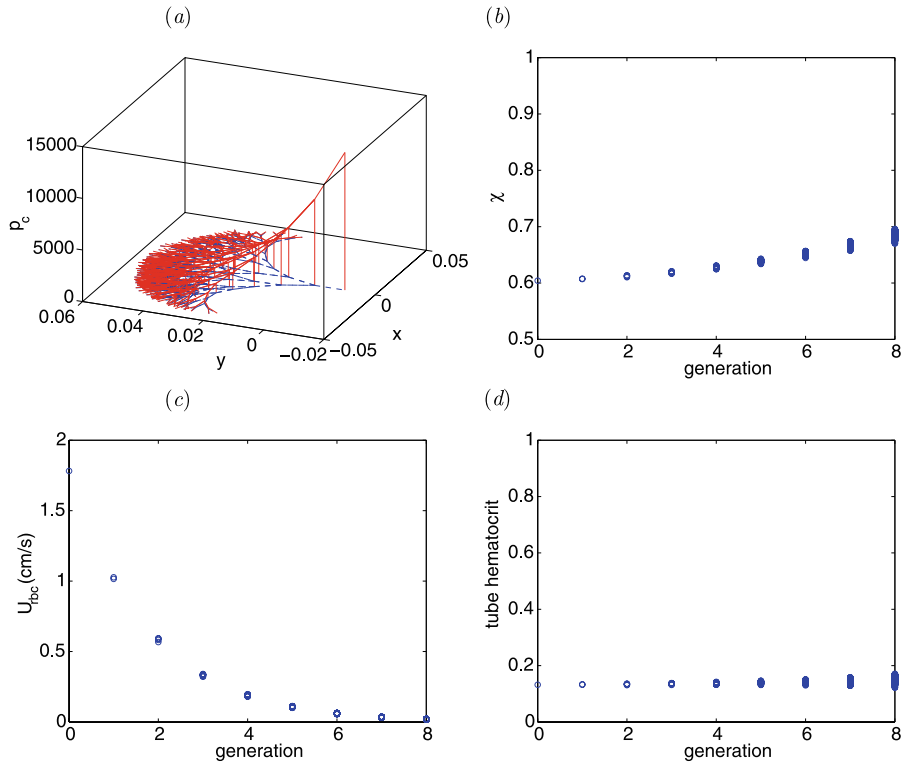


Fig. 13 Same as Fig. 12 but for inlet discharge hematocrit 0.2188.

ocrit 0.4375. In the continuum model, the bifurcation law exponent q is computed as part of the solution and is found to lie in the range between 1.33 and 1.57, which is close to the value of unity used in the discrete model represented by Fig. 6. Figure 12(a) illustrates the distribution of the capillary pressure on the projection of the capillary network in the xy plane, showing an upward concave shape. Figure 12(b) illustrates the distribution of the hematocrit ratio across the network, varying between 0.7 and 0.8. We recall that the discrete cell simulations presented earlier in this paper correspond to $\chi = 0.8$ in all capillary segments. Figure 12(c) illustrates the distribution of the mean cell velocity computed from the mean capillary velocity and hematocrit ratio. The results are consistent with those shown in Fig. 6(c) at steady state. Figure 12(d) illustrates the distribution of the tube hematocrit across the network. Comparing the results to those shown in Fig. 6(g), we note that the discrete model predicts a significantly higher variance across the network due to the random motion of the individual cells at each bifurcation. In contrast, variations in the continuum model rely exclusively on differences in the geometrical structure of the network determined by the random amplitudes, ϵ_L , ϵ_θ , and ϵ_z .

Similar results were obtained for lower hematocrits. Figure 13 shows the predictions of the continuum model for vascular geometry and pressure conditions identical to those corresponding to Fig. 8, and inlet discharge hematocrit 0.2188. The bifurcation law exponent q in the continuum model lies in the range between 1.45 and 1.77. Comparing the

results shown in Fig. 12(d) to those shown in Fig. 8(g), we confirm that the discrete model predicts a significantly higher variance across the network due to the random motion of the individual cells at each bifurcation.

7. Conclusions

We have implemented a numerical method for computing blood flow in a capillary network by pursuing the motion of the individual cells as they enter a network through a single arterial entrance point. Physically, when a cell enters a capillary, it elevates the effective viscosity of the fluid inside the capillary, reduces the flow rate, and causes preceding cells to follow a different path with a higher flow rate. However, the effect is nonlocal and the motion of the cells is felt throughout the entire network. The cell-level approach utilizes information provided directly from cell biofluid dynamics, and thus circumvents the use of correlations and phenomenological approximations. The stochastic input associated with the randomness of the cell motion at a bifurcation distinguishes the present from previous discrete or continuum blood flow models. The hydrodynamics of the network comprises a nonlinear stochastic system forced by the probability of directional cell motion at branching points and bifurcations. We have found that the discrete model predicts a significantly higher variance of the hematocrit across the network due to the randomness of the motion of the individual cells at each bifurcation. In contrast, variations in the continuum model rely exclusively on differences in the geometrical structure of the network.

Our results have indicated that the dependence of the effective viscosity on the tube hematocrit promotes spatial inhomogeneity of the cell distribution. This observation further underlines the importance of the relation between the effective apparent viscosity, the cell velocity, and the tube radius in capillary flow. In the simulations, an approximate relation that mediates a mild dependence was used based on a simplification of numerical results provided by Secomb et al. (1986). Capillary undulations, the noncircular cross-section of the capillaries, and the endothelial surface layer have not been taken into consideration.

We have computed blood flow in an artificially constructed capillary tree similar to that seen in artery and vein networks. Morphometry of actual capillary networks in tissue reveals a more complex structure (Kassab et al., 1993; Kassab and Fung, 1994; Secomb, 2005). For example, capillary branches of Y-, T-, hairpin-, and H-type are found in pig coronary capillary networks (Kassab and Fung, 1994). It is possible that the different topology of these networks may support a different dynamic behavior, including the occurrence of spontaneous oscillations (Carr and Lacoïn, 2000; Carr et al., 2005; Geddes et al., 2007; Kiani et al., 1994). Depending on the physical circumstances and network topology, the cells may act as oscillators instead of adaptive controllers. In future research, the computational methodology presented in this paper will be applied to physiological tissue networks.

The proposed numerical model is a point of departure for computing unsteady oxygen transport from the microvascular network to tissue based on a discrete cell representation. In this model, the tissue is regarded as an isotropic medium with a prescribed oxygen diffusivity and solubility. Oxygen concentration is governed by the unsteady diffusion equation in the presence of homogeneous consumption according to the Michaelis–Menten

kinetics law. Oxygen is delivered to tissue from the blood plasma as well as from the individual red blood cells moving along the capillaries. Red blood cells appear as moving point sources with diminishing intensity acting like Green's functions. The development of this model will be considered in future research.

Acknowledgement

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